Lesson 9: How Much Is Useful?

Unit

Energy: Stores and Transfers

Lesson focus

Students calculate efficiency from energy or power, as a decimal and a percentage, find the useful output from a given efficiency, measure the efficiency of a bouncing ball, and draw and read Sankey diagrams.

Specification

AQA GCSE Physics 4.1.2.2 Efficiency. The specification asks students to recall and apply both efficiency equations; both are also on the AQA Physics equation sheet. Sankey diagrams are not in the specification; they are used here to picture efficiency.

Scope

Students calculate efficiency, and rearrange to find the useful output or the total input.

Learning objectives

By the end of the lesson, students should be able to:

1. Calculate efficiency from energy or power, as a decimal and as a percentage.
2. Explain why efficiency can never be more than 100%, and why most devices are much less efficient.
3. Calculate the useful output or the total input from the efficiency.
4. Draw and interpret a Sankey diagram.
5. Describe ways to increase the efficiency of an energy transfer.

Key vocabulary

Keyword	Student-friendly definition
Useful output	The energy (or power) transferred to the store we want
Wasted output	The energy (or power) dissipated to stores we do not want, usually thermal
Total input	All the energy (or power) supplied
Efficiency	The fraction of the input that is useful
Sankey diagram	A diagram in which the width of each arrow shows the amount of energy

The equations

> *efficiency = useful output energy transfer ÷ total input energy transfer*
> *efficiency = useful power output ÷ total power input*

Efficiency has no unit. As a decimal it is between 0 and 1; multiply by 100 for a percentage.

Rearranged (with efficiency as a decimal): useful output = efficiency × total input, and total input = useful output ÷ efficiency.

Also:

> *total input = useful output + wasted output*

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Equipment and resources

- Per group: two or three different balls, balance, metre rule clamped vertically, a smartphone to film the bounce (optional)
- Sankey-diagram grid in the workbook; rulers, coloured pencils
- Calculators

Starter: Where does the 100 J go?

A lamp is supplied with 100 J. 20 J is transferred by light, 80 J heats the surroundings.

Ask: how much is useful? How much is wasted? Is any destroyed? (20 J; 80 J; none.)

Main teaching: Calculating efficiency

Worked example 1 (energy, as a decimal and a percentage):

> *A motor is supplied with 500 J and transfers 400 J to the kinetic store of the load. Calculate its efficiency.*
> *efficiency = useful output ÷ total input*
> *efficiency = 400 ÷ 500*
> *efficiency = 0.80, or 80%*

Worked example 2 (power):

> *A 2000 W kettle has a useful power output of 1800 W, heating the water. Calculate its efficiency.*
> *efficiency = useful power output ÷ total power input*
> *efficiency = 1800 ÷ 2000*
> *efficiency = 0.90, or 90%*

Worked example 3 (finding the useful output):

> *A 60 W lamp has an efficiency of 25%. Calculate its useful power output.*
> *efficiency = 25% = 0.25*
> *useful output = efficiency × total input*
> *useful output = 0.25 × 60*
> *useful output = 15 W*

Key check: efficiency can never be greater than 1 (100%). If an answer is, the numbers are upside down.

Practical: How efficient is a bounce?

1. Measure the ball's mass with a balance and convert to kilograms.
2. Drop it from 1.00 m, measured from the bottom of the ball.
3. Measure the height of the first bounce (filming against the metre rule helps). Repeat three times; find the mean.
4. Repeat for other balls.

Calculate the two gravitational potential energies separately using E_p = m g h and g = 9.8 N/kg. The initial energy is the input; the energy at the top of the rebound is the useful output. Then divide useful output by input. Keep unrounded energies until the final step.

Sample for a 0.10 kg ball, drop height 1.00 m and mean rebound height 0.65 m:

> *Input $E_{p_1} = 0.10 \times 9.8 \times 1.00 = 0.98$ J*
> *Useful output $E_{p_2} = 0.10 \times 9.8 \times 0.65 = 0.637$ J*
> *efficiency = $E_{p_2} \div E_{p_1} = 0.637 \div 0.98 = 0.65$, or 65%*

Ask: where did the other 35% go? (Thermal store of the ball and floor; sound.) This links back to the shock absorbers of Unit 1, Lesson 3: a good shock absorber is an inefficient bounce.

Main teaching: Sankey diagrams

Draw the lamp from the starter on squared paper: a 100 J arrow 10 squares wide, splitting into a 20 J useful arrow (2 squares) going straight on and an 80 J wasted arrow (8 squares) bending downwards.

Rules:

1. The width of each arrow is proportional to the energy.
2. Useful energy goes straight on; wasted energy bends away (usually down).
3. Input width = useful width + wasted width.

Students then read efficiency from a given Sankey diagram, and draw one for the motor in worked example 1.

Increasing efficiency

Way	Reduces
Lubricate moving parts	Energy transferred by friction to the thermal store
Streamline the shape	Energy transferred to the thermal store of the air by air resistance
Insulate	Energy transferred by heating to the surroundings
Use an LED instead of a filament lamp	Energy transferred by heating to the surroundings

Check for understanding

1. A device transfers 300 J usefully from 1200 J supplied. What is its efficiency? (0.25, or 25%)
2. Why can efficiency never be more than 100%?
3. A 400 W motor is 75% efficient. Calculate its useful power output. (300 W)
4. A lamp is 0.20 efficient and gives 12 J by light. How much energy was supplied? (60 J)
5. On a Sankey diagram, how do you know which arrow is the wasted energy?

Plenary

Students complete: "A device can never be more than 100% efficient, because …" (energy cannot be created, so the useful output cannot be more than the total input). Extension: "Most devices are much less efficient, because …" (some energy is dissipated, usually to the thermal store of the surroundings).

Possible prep

Complete the Sankey diagram page. Stretch: a car engine is about 30% efficient. Draw its Sankey diagram, and suggest two ways to increase its efficiency.

Teacher notes

1. Students put the numbers upside down. Check every answer is at most 1.
2. Students forget to convert a percentage to a decimal before multiplying.

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3 Students give efficiency a unit (J or W). It has none.

4 In the bounce practical, calculate both energies explicitly. Do not introduce a shortcut using a ratio of heights; students practise $E_p = m\ g\ h$ and then the standard efficiency equation.

5 Week 5's second lesson is set aside for revision and catch-up before the test in week 6.

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